Linear transformation on $\mathbb{R}^n$

Goals

When a matrix A multiplies a vector $\mathbf{v}$, it "transforms" $\mathbf{v}$ into another vector $A\mathbf{v}$. ***In goes v, out comes*** $T(\boldsymbol{v}) = A\boldsymbol{v}$. A transformation T follows the same ideas as a function

1. Linear transformation $T : \mathbb{R}^m \to \mathbb{R}^n$
2. Matrix transformation
3. Some special linear transformation: reflection, rotation and projection

Table of contents

1. Linear Transformation
   - Linear Transformation
   - Geometric Linear Transform of $\mathbb{R}^2$
2. Kernel and Image of Linear Transformation
   - One-to-one and Onto Transformation
   - Isomorphism and Composition
   - Kernel and Image

Linear transformation

A function

$$T: \ \mathbb{R}^m \to \mathbb{R}^n$$
$$\mathbf{u} \mapsto T(\mathbf{u}) \text{ (called } \textbf{image} \text{ of } \mathbf{u})$$

is a **linear transformation** if

1. $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}+T(\mathbf{v}))$ for all $\mathbf{u}, \mathbf{v} \in U$
2. $T(c\mathbf{u}) = cT(\mathbf{u})$ for any vector $\mathbf{u} \in U$ and any real number c

Example

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be defined by

$$T\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) = \begin{bmatrix} u_1 + 1 \\ 2u_2 \\ u_3 \end{bmatrix}$$

To determine whether T is a linear transformation, let

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

then

$$T(\mathbf{u} + \mathbf{v}) = T\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) = T\left(\begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{bmatrix}\right) = \begin{bmatrix} u_1 + v_1 + 1 \\ 2(u_2 + v_2) \\ u_3 \end{bmatrix}$$

$$T(\mathbf{u}) = \begin{bmatrix} u_1 + 1 \\ 2u_2 \\ u_3 \end{bmatrix} \text{ and } T(\mathbf{v}) = \begin{bmatrix} v_1 + 1 \\ 2v_2 \\ v_3 \end{bmatrix} \Longrightarrow T(\mathbf{u}) + T(\mathbf{v}) = \begin{bmatrix} u_1 + v_1 + 2 \\ 2(u_2 + v_2) \\ u_3 + v_3 \end{bmatrix}$$

It is clear that $T(\mathbf{u}) + T(\mathbf{v})$ is not always equal to $T(\mathbf{u} + \mathbf{v})$. So T is not a linear transformation.

Example

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by

$$T\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) = \begin{bmatrix} u_1 + 2u_2 \\ 3u_2 - 2u_3 \end{bmatrix}$$

Show that T is a linear transformation

Solution

Let

$$\boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \text{ and } \boldsymbol{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

1. $T(\boldsymbol{u}+\boldsymbol{v}) = T(\boldsymbol{u}) + T(\boldsymbol{v}) = \begin{bmatrix} u_1 + v_1 + 2(u_2 + v_2) \\ 3(u_2 + v_2) - 2(u_3 + v_3) \end{bmatrix}$
2. $T(c\boldsymbol{u}) = cT(\boldsymbol{u}) = \begin{bmatrix} c(u_1 + 2u_2) \\ c(3u_2 - 2u_3) \end{bmatrix}$

T satisfies two conditions of linear transformation. So T is a linear transformation.

Theorem (Linearity Theorem)

If $T : \mathbb{R}^n \to \mathbb{R}^m$ *is a linear transformation then*

$$T(a_1\boldsymbol{x}_1 + a_2\boldsymbol{x}_2 + \cdots + a_n\boldsymbol{x}_n) = a_1T(\boldsymbol{x}_1) + a_2T(\boldsymbol{x}_2) + \cdots + a_nT(\boldsymbol{x}_n)$$

for all scalar a_i *and vector* $\boldsymbol{x}_i \in \mathbb{R}^n$.

Problem

Let $T : \mathbb{R}^3 \to \mathbb{R}^4$ be a linear transformation such that

$$T\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 0 \\ -2 \end{bmatrix} \text{ and } T\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ -1 \\ 5 \end{bmatrix}.$$

Problem

Let $T : \mathbb{R}^3 \to \mathbb{R}^4$ be a linear transformation such that

$$T\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 0 \\ -2 \end{bmatrix} \text{ and } T\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ -1 \\ 5 \end{bmatrix}. \text{ Find } T\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix}.$$

Problem

Let $T : \mathbb{R}^3 \to \mathbb{R}^4$ be a linear transformation such that

$$T\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 0 \\ -2 \end{bmatrix} \text{ and } T\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ -1 \\ 5 \end{bmatrix}. \text{ Find } T\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix}.$$

Solution

Problem

Let $T : \mathbb{R}^3 \to \mathbb{R}^4$ be a linear transformation such that

$$T\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 0 \\ -2 \end{bmatrix} \text{ and } T\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ -1 \\ 5 \end{bmatrix}. \text{ Find } T\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix}.$$

Solution

The only way it is possible to solve this problem is if

$$\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix} \text{ is a linear combination of } \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix},$$

Problem

Let $T : \mathbb{R}^3 \to \mathbb{R}^4$ be a linear transformation such that

$$T\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 0 \\ -2 \end{bmatrix} \text{ and } T\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ -1 \\ 5 \end{bmatrix}. \text{ Find } T\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix}.$$

Solution

The only way it is possible to solve this problem is if

$$\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix} \text{ is a linear combination of } \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix},$$

i.e., if there exist $a, b \in \mathbb{R}$ so that

$$\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix} = a\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} + b\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}.$$

Solution (continued)

To find a and b, solve the system of three equations in two variables:

$$\left[\begin{array}{cc|r} 1 & 4 & -7 \\ 3 & 0 & 3 \\ 1 & 5 & -9 \end{array}\right] \to \cdots \to \left[\begin{array}{cc|r} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right]$$

Thus $a = 1$, $b = -2$, and

$$\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}.$$

Solution (continued)

We now use that fact that linear transformations preserve linear combinations, implying that

$$T\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix} = T\left(\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} - 2\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}\right)$$

Solution (continued)

We now use that fact that linear transformations preserve linear combinations, implying that

$$\begin{aligned} T\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix} &= T\left(\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} - 2\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}\right) \\ &= T\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} - 2T\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix} \end{aligned}$$

Solution (continued)

We now use that fact that linear transformations preserve linear combinations, implying that

$$
\begin{aligned}
T\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix} &= T\left(\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} - 2\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}\right) \\
&= T\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} - 2T\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix} \\
&= \begin{bmatrix} 4 \\ 4 \\ 0 \\ -2 \end{bmatrix} - 2\begin{bmatrix} 4 \\ 5 \\ -1 \\ 5 \end{bmatrix} = \begin{bmatrix} -4 \\ -6 \\ 2 \\ -12 \end{bmatrix}
\end{aligned}
$$

Solution (continued)

We now use that fact that linear transformations preserve linear combinations, implying that

$$\begin{aligned} T\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix} &= T\left(\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} - 2\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}\right) \\ &= T\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} - 2T\begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix} \\ &= \begin{bmatrix} 4 \\ 4 \\ 0 \\ -2 \end{bmatrix} - 2\begin{bmatrix} 4 \\ 5 \\ -1 \\ 5 \end{bmatrix} = \begin{bmatrix} -4 \\ -6 \\ 2 \\ -12 \end{bmatrix} \end{aligned}$$

Therefore, $T\begin{bmatrix} -7 \\ 3 \\ -9 \end{bmatrix} = \begin{bmatrix} -4 \\ -6 \\ 2 \\ -12 \end{bmatrix}$.

Example

If $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear transformation with $T\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}2\\-3\end{bmatrix}$ and $T\begin{bmatrix}1\\-2\end{bmatrix} = \begin{bmatrix}5\\1\end{bmatrix}$.

Find $T\begin{bmatrix}4\\3\end{bmatrix}$

Solution

Write $\boldsymbol{z} = \begin{bmatrix}4\\3\end{bmatrix}$, $\boldsymbol{x} = \begin{bmatrix}1\\1\end{bmatrix}$ *and* $\boldsymbol{y} = \begin{bmatrix}1\\-2\end{bmatrix}$ *then*

$$\boldsymbol{z} = \frac{11}{3}\boldsymbol{x} + \frac{1}{3}\boldsymbol{y}$$

So

$$T(\boldsymbol{z}) = \frac{11}{3}T(\boldsymbol{x}) + \frac{1}{3}T(\boldsymbol{y}) = \frac{11}{3}\begin{bmatrix}2\\-3\end{bmatrix} + \frac{1}{3}\begin{bmatrix}5\\1\end{bmatrix} = \frac{1}{3}\begin{bmatrix}27\\-32\end{bmatrix}$$

Exercise

Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be a linear transformation such that

$$T\begin{bmatrix} 1 \\ 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \text{ and } T\begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}. \text{ Find } T\begin{bmatrix} 2 \\ 5 \\ -3 \\ -7 \end{bmatrix}.$$

Matrix transformation

If A is an $m \times n$ matrix, the transformation

$$T_A : \mathbb{R}^n \to \mathbb{R}^m$$

defined by

$$T_A(\mathbf{x}) = A\mathbf{x} \qquad \forall \mathbf{x} \in \mathbb{R}^n$$

is called the matrix transformation induced by A

Think of the matrix A as an object that 'acts' on a vector $\mathbf{x}$ by multiplication to produce a new vector called $A\mathbf{x}$

Some Special Matrix Transformations

Example (The Zero Transformation)

If A is the $m \times n$ matrix of zeros, then the transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ induced by A is called the zero transformation because for every vector $\mathbf{x}$ in $\mathbb{R}^n$

$$T(\mathbf{x}) = A\mathbf{x} = 0\mathbf{x} = \mathbf{0}.$$

Some Special Matrix Transformations

Example (The Zero Transformation)

If A is the $m \times n$ matrix of zeros, then the transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ induced by A is called the zero transformation because for every vector $\mathbf{x}$ in $\mathbb{R}^n$

$$T(\mathbf{x}) = A\mathbf{x} = 0\mathbf{x} = \mathbf{0}.$$

Note that the first zero is the matrix A, while the second zero is the zero vector of $\mathbb{R}^m$.

The zero transformation is usually written as $T = 0$.

Some Special Matrix Transformations

Example (The Zero Transformation)

If A is the $m \times n$ matrix of zeros, then the transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ induced by A is called the zero transformation because for every vector $\mathbf{x}$ in $\mathbb{R}^n$

$$T(\mathbf{x}) = A\mathbf{x} = 0\mathbf{x} = \mathbf{0}.$$

Note that the first zero is the matrix A, while the second zero is the zero vector of $\mathbb{R}^m$.
The zero transformation is usually written as $T = 0$.

Example (The Identity Transformation)

The transformation of $\mathbb{R}^n$ induced by I_n, the $n \times n$ identity matrix, is called the identity transformation because for every vector $\mathbf{x}$ in $\mathbb{R}^n$,

$$T(\mathbf{x}) = \mathbf{I}_n\mathbf{x} = \mathbf{x}.$$

The identity transformation on $\mathbb{R}^n$ is usually written as $1_{\mathbb{R}^n}$.

Example (reflection about x - axis)

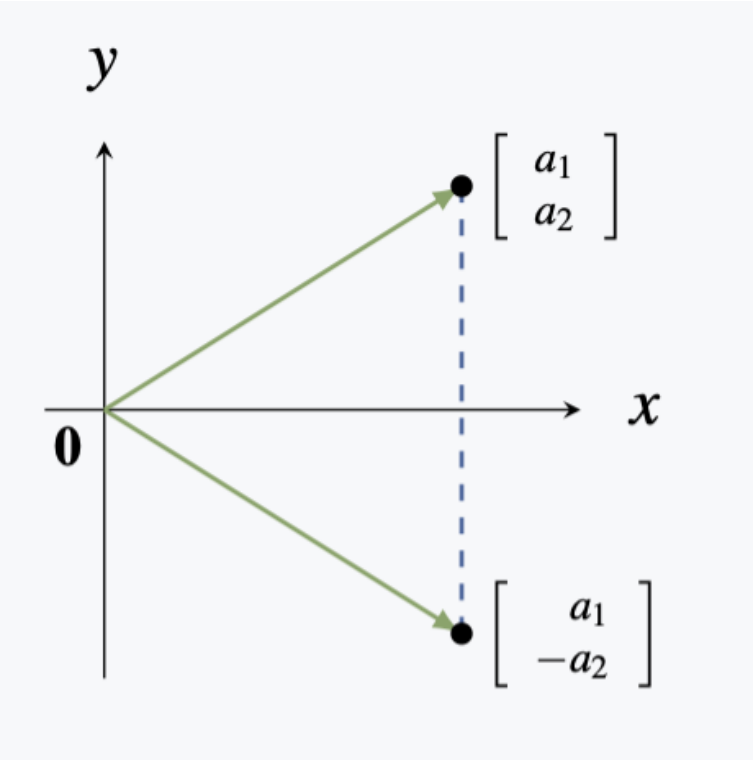
Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

For any $\mathbf{x} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$, we have

$$T_A(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} a_1 \\ -a_2 \end{bmatrix}$$

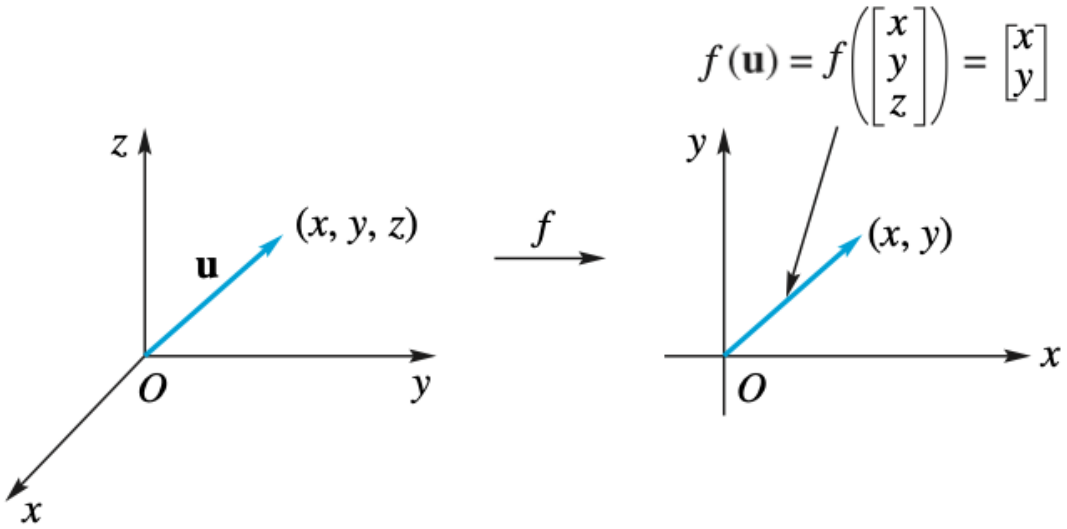
This transformation carries a vector to its reflection about x - axis



Example (projection)

Let $f : \mathbb{R}^3 \to \mathbb{R}^2$ be the matrix transformation defined by

$$f(\mathbf{u}) = f\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$



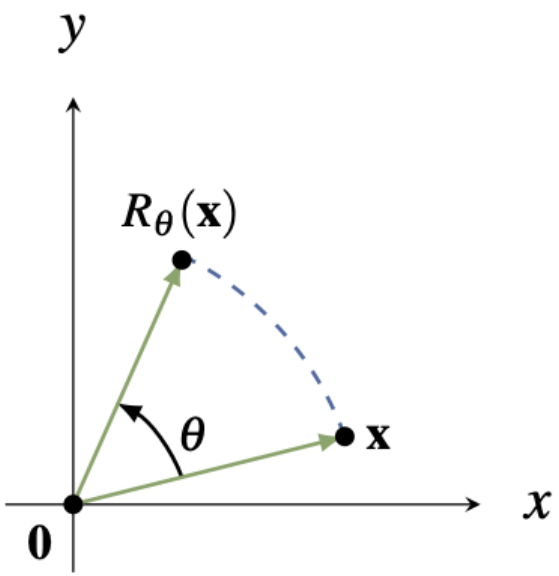
The effect of this matrix transformation, which is called projection into the xy-plane.

Example (rotation)

Given an angle θ, the matrix transformation $R_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ corresponding to the matrix

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

has the following action: counterclockwise rotation of $\mathbb{R}^2$ about the origin through the angle θ



Example (Revisited)

Let $T : \mathbb{R}^3 \to \mathbb{R}^4$ be defined by

$$T\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a+b \\ b+c \\ a-c \\ c-b \end{bmatrix}$$

Example (Revisited)

Let $T : \mathbb{R}^3 \to \mathbb{R}^4$ be defined by

$$T\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a+b \\ b+c \\ a-c \\ c-b \end{bmatrix}$$

Is T a matrix transformation?

Example (Revisited)

Let $T : \mathbb{R}^3 \to \mathbb{R}^4$ be defined by

$$T\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a+b \\ b+c \\ a-c \\ c-b \end{bmatrix}$$

Is T a matrix transformation?

Consider $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \end{bmatrix}$, then

$$A\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a+b \\ b+c \\ a-c \\ c-b \end{bmatrix} = T\begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

Example (Revisited)

Let $T : \mathbb{R}^3 \to \mathbb{R}^4$ be defined by

$$T\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a+b \\ b+c \\ a-c \\ c-b \end{bmatrix}$$

Is T a matrix transformation?

Consider $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \end{bmatrix}$, then

$$A\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a+b \\ b+c \\ a-c \\ c-b \end{bmatrix} = T\begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

So in this case T is a matrix transformation!

Not all transformations are matrix transformations!

Example

Consider $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \mathbf{x} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ for all } \mathbf{x} \in \mathbb{R}^2.$$

Not all transformations are matrix transformations!

Example

Consider $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \mathbf{x} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ for all } \mathbf{x} \in \mathbb{R}^2.$$

Why is T not a matrix transformation?

Example (continued)

We have $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \mathbf{x} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ for all } \mathbf{x} \in \mathbb{R}^2.$$

Example (continued)

We have $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \mathbf{x} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ for all } \mathbf{x} \in \mathbb{R}^2.$$

Since every matrix transformation is a linear transformation,

Example (continued)

We have $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \mathbf{x} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ for all } \mathbf{x} \in \mathbb{R}^2.$$

Since every matrix transformation is a linear transformation, we consider $T(\vec{0}$, where $\vec{0}$ is the zero vector of $\mathbb{R}^2$.

Example (continued)

We have $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \mathbf{x} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ for all } \mathbf{x} \in \mathbb{R}^2.$$

Since every matrix transformation is a linear transformation, we consider $T(\vec{0}$, where $\vec{0}$ is the zero vector of $\mathbb{R}^2$.

$$T \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Example (continued)

We have $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \mathbf{x} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ for all } \mathbf{x} \in \mathbb{R}^2.$$

Since every matrix transformation is a linear transformation, we consider $T(\vec{0}$, where $\vec{0}$ is the zero vector of $\mathbb{R}^2$.

$$T\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Example (continued)

We have $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \mathbf{x} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ for all } \mathbf{x} \in \mathbb{R}^2.$$

Since every matrix transformation is a linear transformation, we consider $T(\vec{0}$, where $\vec{0}$ is the zero vector of $\mathbb{R}^2$.

$$T\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Example (continued)

We have $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \mathbf{x} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ for all } \mathbf{x} \in \mathbb{R}^2.$$

Since every matrix transformation is a linear transformation, we consider $T(\vec{0}$, where $\vec{0}$ is the zero vector of $\mathbb{R}^2$.

$$T\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

violating one of the properties of a linear transformation.

Example (continued)

We have $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \mathbf{x} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ for all } \mathbf{x} \in \mathbb{R}^2.$$

Since every matrix transformation is a linear transformation, we consider $T(\vec{0}$, where $\vec{0}$ is the zero vector of $\mathbb{R}^2$.

$$T\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

violating one of the properties of a linear transformation.

Therefore, T is not a linear transformation, and hence is not a matrix transformation.
Can you see any other reasons why T is not a matrix transformation?

Matrix transformation is a linear transformation $\mathbb{R}^n \to \mathbb{R}^m$

Proof.

We need to verify 2 properties of linear transformation

1. $L(\mathbf{u}+\mathbf{v}) = A(\mathbf{u}+\mathbf{v}) = A\mathbf{u} + A\mathbf{v} = L(\mathbf{u}) + L(\mathbf{v})$
2. $L(c\mathbf{u}) = A(c\mathbf{u}) = cA(\mathbf{u}) = cL(\mathbf{u})$

Hence every matrix transformation is a linear transformation. $\square$

Linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is a matrix transformation

Let $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ be the standard basis of $\mathbb{R}^n$. For every $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \ldots \\ x_n \end{bmatrix} \in \mathbb{R}^n$, we have $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n$

So

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + x_2T(\mathbf{e}_2) + \cdots + x_nT(\mathbf{e}_n)$$

$$= \underbrace{\begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \ldots & T(\mathbf{e}_n) \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \\ \ldots \\ x_n \end{bmatrix} = A\mathbf{x}$$

T is the matrix transformation induced by matrix A

Example

Let

$$T: \ \mathbb{R}^3 \to \mathbb{R}^3$$
$$\mathbf{x} \mapsto r\mathbf{x}.$$

T is called **dilation** if $r > 1$ and **contraction** if $r < 1$.
Prove that T is a linear transformation and find the associated transformation matrix A.

Solution (1) - Prove that T is a linear transformation

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$

1. $$T(\mathbf{u}+\mathbf{v}) = r(\mathbf{u}+\mathbf{v})$$

   and

   $$T(\mathbf{u}) + T(\mathbf{v}) = r\mathbf{u} + r\mathbf{v}$$

   so $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
2. For any $c \in \mathbb{R}$, we have

   $$T(c\mathbf{u}) = r(c\mathbf{u}) = (cr)\mathbf{u}$$

   and

   $$cT(\mathbf{u}) = c(r\mathbf{u}) = (cr)\mathbf{u}$$

   So $T(c\mathbf{u}) = cT(\mathbf{u})$

Hence T is a linear transformation.

Solution (2) - Find matrix A

Let $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ be the standard basis of $\mathbb{R}^3$. We now compute $T(\mathbf{e}_i)$

$$T(\mathbf{e}_1) = r\mathbf{e}_1 = \begin{bmatrix} r \\ 0 \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_2) = r\mathbf{e}_2 = \begin{bmatrix} 0 \\ r \\ 0 \end{bmatrix}, \quad T(\mathbf{e}_3) = r\mathbf{e}_3 \begin{bmatrix} 0 \\ 0 \\ r \end{bmatrix}$$

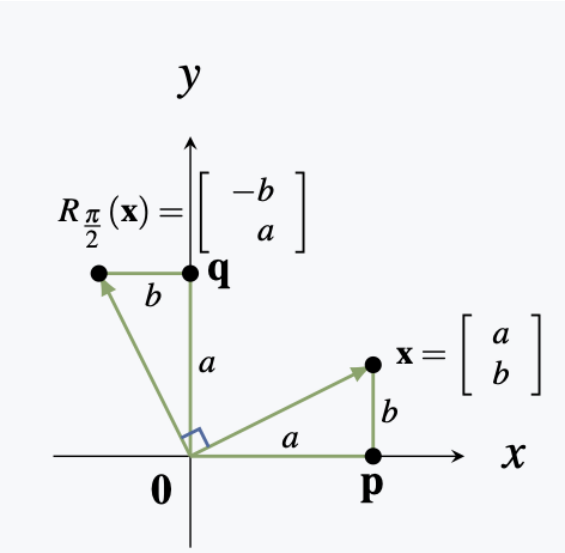
Hence $A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & T(\mathbf{e}_3) \end{bmatrix} = \begin{bmatrix} r & 0 & 0 \\ 0 & r & 0 \\ 0 & 0 & r \end{bmatrix}$

Example - Rotation

$$R_{\frac{\pi}{2}} : \mathbb{R}^2 \to \mathbb{R}^2$$

denote counterclockwise rotation about the origin through $\frac{\pi}{2}$ radians (that is, 90°).
Then

$$R_{\frac{\pi}{2}} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -b \\ a \end{bmatrix}.$$



It is easy to verify that $R_{\frac{\pi}{2}}$ is a linear transformation.

Because $R_{\frac{\pi}{2}}(\mathbf{e}_1) = R_{\frac{\pi}{2}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $R_{\frac{\pi}{2}}(\mathbf{e}_2) = R_{\frac{\pi}{2}} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$, the transformation matrix of $R_{\frac{\pi}{2}}$ is

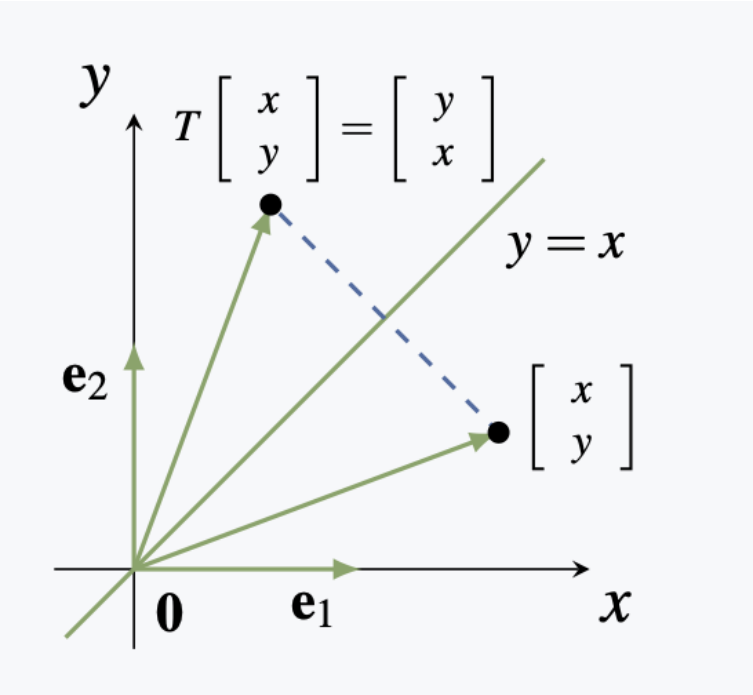
$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Exercise

Let

$$T: \ \mathbb{R}^2 \to \mathbb{R}^2$$
$$\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} y \\ x \end{bmatrix}$$

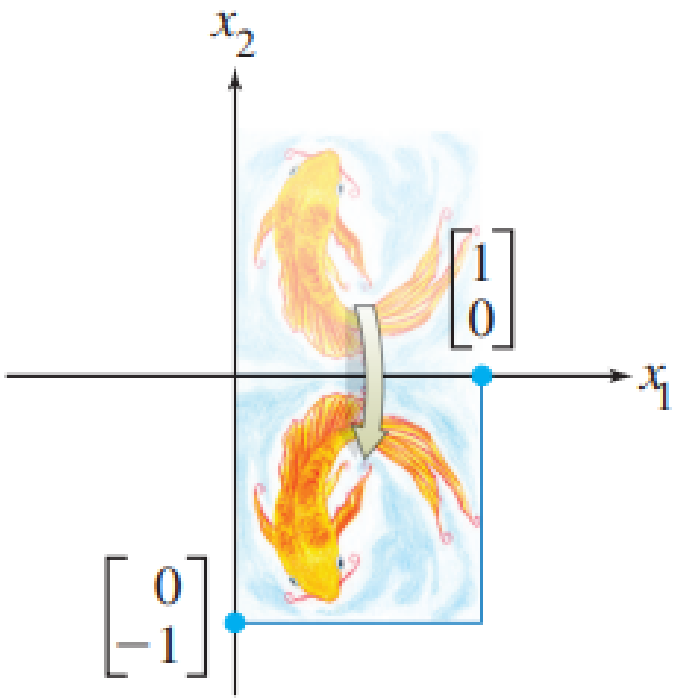
be the reflection in the line $y = x$.



Prove that T is a linear transformation and find its matrix.

Reflection (1)

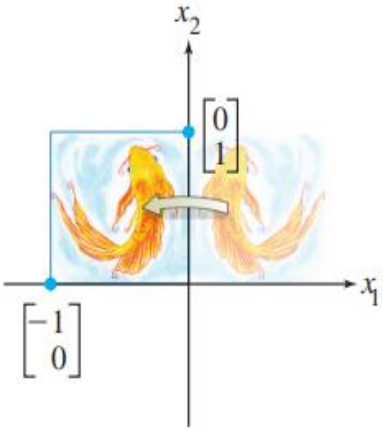
Reflection through the x_1-axis



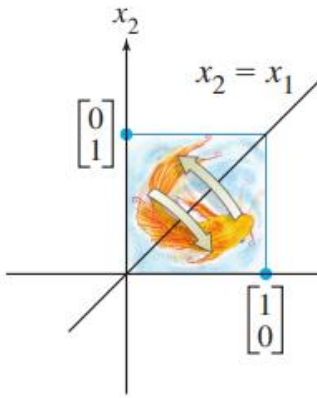
The inducing matrix is $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Reflection (2)

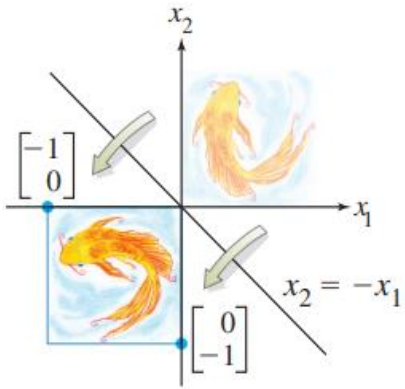
Reflection through the x_2-axis



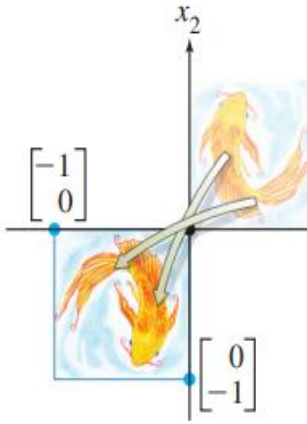
Reflection through the line $x_2 = x_1$



Reflection through the line $x_2 = -x_1$

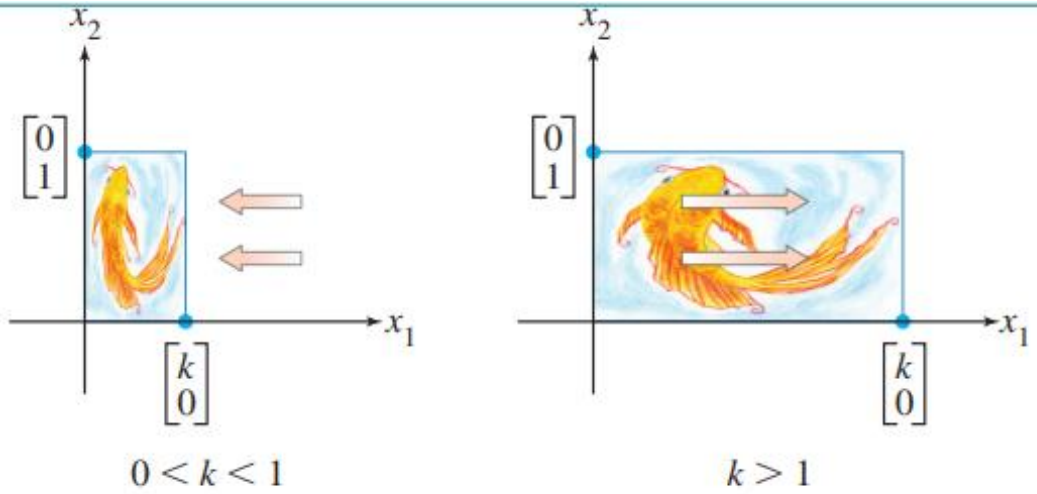


Reflection through the origin



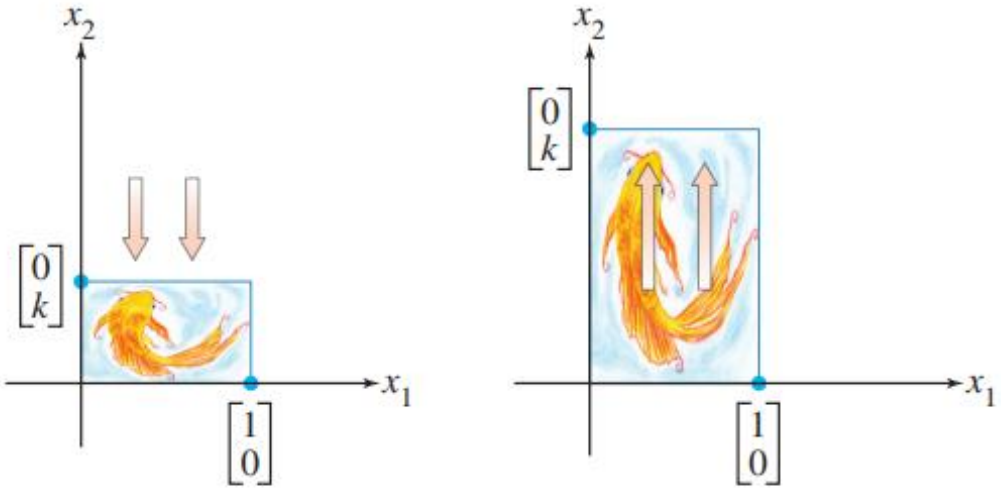
Contraction and Expansion

Horizontal contraction and expansion



$$\begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$$

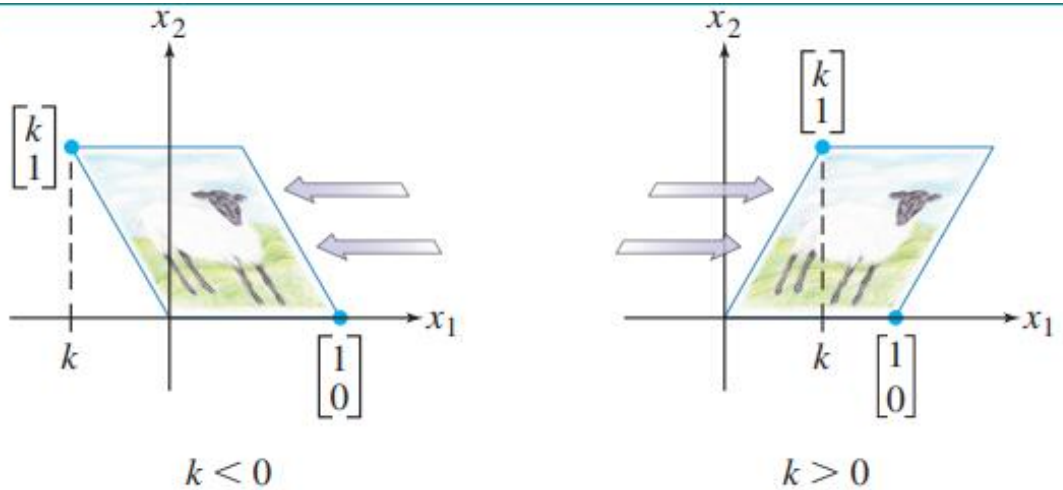
Vertical contraction and expansion



$$\begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$$

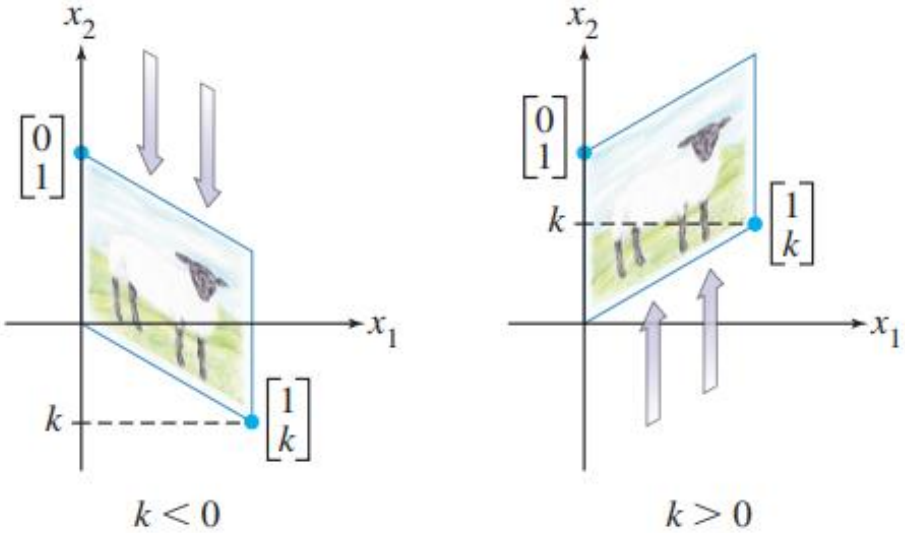
Shears

Horizontal shear



$$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

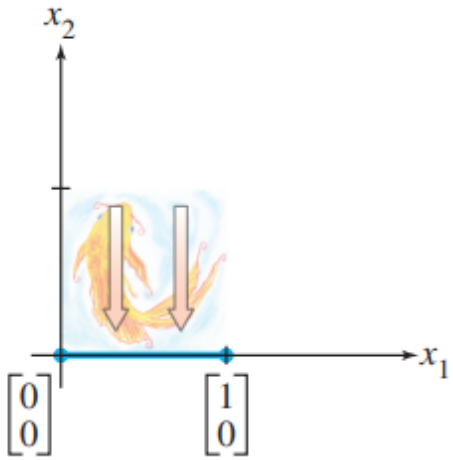
Vertical shear



$$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

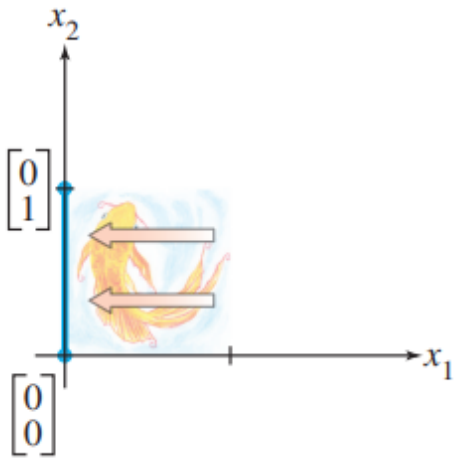
Projections

Projection onto the x_1-axis



$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Projection onto the x_2-axis



$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Find the Matrix of a Linear Transform T

Problem

Sometimes T is not defined so nicely for us. Suppose a linear transform T is given as

$$T\begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \; T\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Find the matrix A of T.

Find the Matrix of a Linear Transform T

Problem

Sometimes T is not defined so nicely for us. Suppose a linear transform T is given as

$$T\begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \; T\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Find the matrix A of T.

Solution

We need to write $\mathbf{e}_1$ and $\mathbf{e}_2$ as a linear combination of the vectors provided. So we reduce the augmented matrix having $\mathbf{e}_1$ and $\mathbf{e}_2$ as the third and fourth columns:

$$\left[\begin{array}{cc|cc} 1 & 1 & \vec{e}_1 & \vec{e}_2 \\ 5 & 4 & & \end{array}\right] = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 5 & 4 & 0 & 1 \end{array}\right]$$

Find the Matrix of a Linear Transform T

Problem

Sometimes T is not defined so nicely for us. Suppose a linear transform T is given as

$$T\begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \; T\begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Find the matrix A of T.

Solution

We need to write $\mathbf{e}_1$ and $\mathbf{e}_2$ as a linear combination of the vectors provided. So we reduce the augmented matrix having $\mathbf{e}_1$ and $\mathbf{e}_2$ as the third and fourth columns:

$$\left[\begin{array}{cc|cc} 1 & 1 & \vec{e}_1 & \vec{e}_2 \\ 5 & 4 & & \end{array}\right] = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 5 & 4 & 0 & 1 \end{array}\right] \to \cdots \to \left[\begin{array}{cc|cc} 1 & 0 & -4 & 1 \\ 0 & 1 & 5 & -1 \end{array}\right]$$

Solution (continued)

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 5 & 4 & 0 & 1 \end{array}\right] \to \cdots \to \left[\begin{array}{cc|cc} 1 & 0 & -4 & 1 \\ 0 & 1 & 5 & -1 \end{array}\right]$$

From this we can see that

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -4 \begin{bmatrix} 1 \\ 5 \end{bmatrix} + 5 \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

and

$$\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix} - \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

Solution (continued)

So

$$T(\mathbf{e}_1) = T\left(-4\begin{bmatrix}1\\5\end{bmatrix} + 5\begin{bmatrix}1\\4\end{bmatrix}\right)$$

Solution (continued)

So

$$T(\mathbf{e}_1) = T\left(-4\begin{bmatrix}1\\5\end{bmatrix} + 5\begin{bmatrix}1\\4\end{bmatrix}\right)$$

$$= -4T\begin{bmatrix}1\\5\end{bmatrix} + 5T\begin{bmatrix}1\\4\end{bmatrix}$$

Solution (continued)

So

$$T(\mathbf{e}_1) = T\left(-4\begin{bmatrix}1\\5\end{bmatrix} + 5\begin{bmatrix}1\\4\end{bmatrix}\right)$$

$$= -4T\begin{bmatrix}1\\5\end{bmatrix} + 5T\begin{bmatrix}1\\4\end{bmatrix}$$

$$= -4\begin{bmatrix}1\\2\end{bmatrix} + 5\begin{bmatrix}3\\2\end{bmatrix}$$

Solution (continued)

So

$$
\begin{aligned}
T(\mathbf{e}_1) &= T\left(-4\begin{bmatrix}1\\5\end{bmatrix} + 5\begin{bmatrix}1\\4\end{bmatrix}\right) \\
&= -4T\begin{bmatrix}1\\5\end{bmatrix} + 5T\begin{bmatrix}1\\4\end{bmatrix} \\
&= -4\begin{bmatrix}1\\2\end{bmatrix} + 5\begin{bmatrix}3\\2\end{bmatrix} \\
&= \begin{bmatrix}11\\2\end{bmatrix}
\end{aligned}
$$

Solution (continued)

So

$$T(\mathbf{e}_1) = T\left(-4\begin{bmatrix}1\\5\end{bmatrix} + 5\begin{bmatrix}1\\4\end{bmatrix}\right)$$

$$= -4T\begin{bmatrix}1\\5\end{bmatrix} + 5T\begin{bmatrix}1\\4\end{bmatrix}$$

$$= -4\begin{bmatrix}1\\2\end{bmatrix} + 5\begin{bmatrix}3\\2\end{bmatrix}$$

$$= \begin{bmatrix}11\\2\end{bmatrix}$$

and

$$T(\mathbf{e}_2) = T\begin{bmatrix}0\\1\end{bmatrix}$$

$$= T\left(\begin{bmatrix}1\\5\end{bmatrix} - \begin{bmatrix}1\\4\end{bmatrix}\right)$$

Solution (continued)

So

$$T(\mathbf{e}_1) = T\left(-4\begin{bmatrix}1\\5\end{bmatrix} + 5\begin{bmatrix}1\\4\end{bmatrix}\right)$$

$$= -4T\begin{bmatrix}1\\5\end{bmatrix} + 5T\begin{bmatrix}1\\4\end{bmatrix}$$

$$= -4\begin{bmatrix}1\\2\end{bmatrix} + 5\begin{bmatrix}3\\2\end{bmatrix}$$

$$= \begin{bmatrix}11\\2\end{bmatrix}$$

and

$$T(\mathbf{e}_2) = T\begin{bmatrix}0\\1\end{bmatrix}$$

$$= T\left(\begin{bmatrix}1\\5\end{bmatrix} - \begin{bmatrix}1\\4\end{bmatrix}\right)$$

$$= T\begin{bmatrix}1\\5\end{bmatrix} - T\begin{bmatrix}1\\4\end{bmatrix}$$

Solution (continued)

So

$$T(\mathbf{e}_1) = T\left(-4\begin{bmatrix}1\\5\end{bmatrix} + 5\begin{bmatrix}1\\4\end{bmatrix}\right)$$

$$= -4T\begin{bmatrix}1\\5\end{bmatrix} + 5T\begin{bmatrix}1\\4\end{bmatrix}$$

$$= -4\begin{bmatrix}1\\2\end{bmatrix} + 5\begin{bmatrix}3\\2\end{bmatrix}$$

$$= \begin{bmatrix}11\\2\end{bmatrix}$$

and

$$T(\mathbf{e}_2) = T\begin{bmatrix}0\\1\end{bmatrix}$$

$$= T\left(\begin{bmatrix}1\\5\end{bmatrix} - \begin{bmatrix}1\\4\end{bmatrix}\right)$$

$$= T\begin{bmatrix}1\\5\end{bmatrix} - T\begin{bmatrix}1\\4\end{bmatrix}$$

$$= \begin{bmatrix}1\\2\end{bmatrix} - \begin{bmatrix}3\\2\end{bmatrix}$$

Solution (continued)

So

$$T(\mathbf{e}_1) = T\left(-4\begin{bmatrix}1\\5\end{bmatrix} + 5\begin{bmatrix}1\\4\end{bmatrix}\right)$$
$$= -4T\begin{bmatrix}1\\5\end{bmatrix} + 5T\begin{bmatrix}1\\4\end{bmatrix}$$
$$= -4\begin{bmatrix}1\\2\end{bmatrix} + 5\begin{bmatrix}3\\2\end{bmatrix}$$
$$= \begin{bmatrix}11\\2\end{bmatrix}$$

and

$$T(\mathbf{e}_2) = T\begin{bmatrix}0\\1\end{bmatrix}$$
$$= T\left(\begin{bmatrix}1\\5\end{bmatrix} - \begin{bmatrix}1\\4\end{bmatrix}\right)$$
$$= T\begin{bmatrix}1\\5\end{bmatrix} - T\begin{bmatrix}1\\4\end{bmatrix}$$
$$= \begin{bmatrix}1\\2\end{bmatrix} - \begin{bmatrix}3\\2\end{bmatrix}$$
$$= \begin{bmatrix}-2\\0\end{bmatrix}$$

Solution (continued)

The matrix of T is the matrix whose first column is $T(\mathbf{e}_1)$, and second column is $T(\mathbf{e}_2)$:

Solution (continued)

The matrix of T is the matrix whose first column is $T(\mathbf{e}_1)$, and second column is $T(\mathbf{e}_2)$:

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) \end{bmatrix} = \begin{bmatrix} 11 & -2 \\ 2 & 0 \end{bmatrix}$$

Table of contents

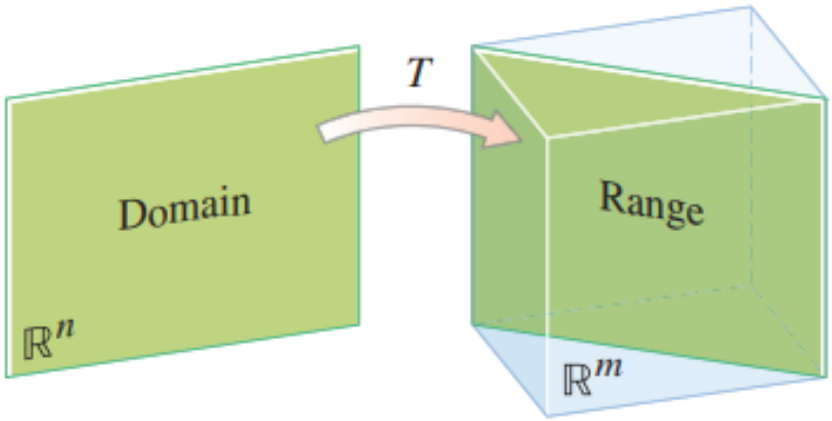
1 Linear Transformation
- Linear Transformation
- Geometric Linear Transform of $\mathbb{R}^2$

2 Kernel and Image of Linear Transformation
- One-to-one and Onto Transformation
- Isomorphism and Composition
- Kernel and Image

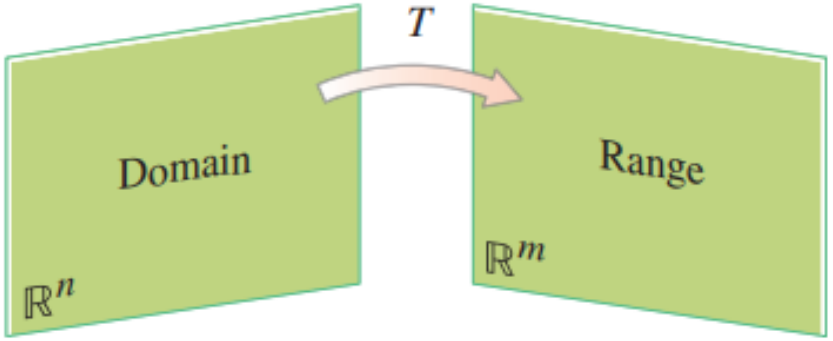
Onto Transformation

Definition

A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is said to be **onto** $\mathbb{R}^m$ if for each $\mathbf{b} \in \mathbb{R}^m$ is an image of *at least one* $\mathbf{x} \in \mathbb{R}^n$



T is *not* onto $\mathbb{R}^m$



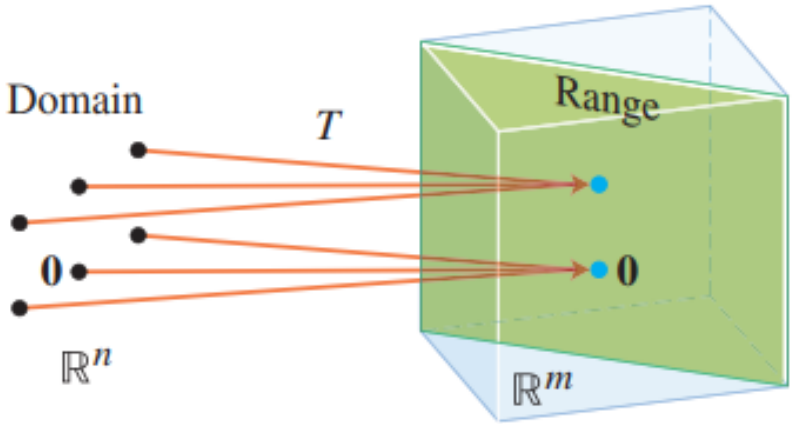
T is onto $\mathbb{R}^m$

One-to-one Transformation

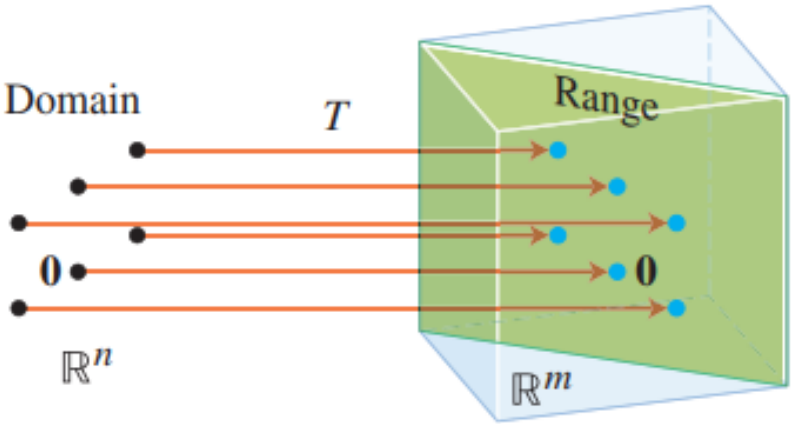
Definition

A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is said to be **one-to-one** if for each $\mathbf{b} \in \mathbb{R}^m$ is an image of *at most one* $\mathbf{x} \in \mathbb{R}^n$.

The equivalent condition for one-to-one is that $T(\mathbf{x}) \neq T(\mathbf{y})$ for all $\mathbf{x} \neq \mathbf{y}$



T is *not* one-to-one



T is one-to-one

Theorem

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation induced by matrix A

1. *T is onto if and only if the columns of A span $\mathbb{R}^m$*
2. *T is one-to-one if and only if $T(\boldsymbol{x}) = 0$ has only trivial solution*
3. *T is one-to-one if and only if the columns of A are linearly independent.*

Proof

1. T is onto if and only for all $\mathbf{b} \in \mathbb{R}^m$, there exists at least $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{b}$. Hence $\mathbf{b} \in col(A)$. Thus $\mathbb{R}^m \subseteq col(A)$. It is equivelent to $\mathbb{R}^m = col(A)$ because $col(A) \subseteq \mathbb{R}^m$
2. If T is one-to-one then $T(\mathbf{x}) = \mathbf{0}$ has at most one solution. It is obvious that $T(\mathbf{0}) = \mathbf{0}$ hence $\mathbf{0}$ is the only trivial solution

   Inversely, if T has only trivial solution then, for $\mathbf{x} \neq \mathbf{y}$, we have $\mathbf{x} - \mathbf{y} \neq \mathbf{0}$ and hence $T(\mathbf{x} - \mathbf{y}) \neq \mathbf{0}$. Hence $T(\mathbf{x}) - T(\mathbf{y}) \neq \mathbf{0}$ and then $T(\mathbf{x}) \neq T(\mathbf{y})$
3. The equation $T(\mathbf{x}) = \mathbf{0}$ is the same as $A\mathbf{x} = \mathbf{0}$. Thus T is one-to-one if and only if $A\mathbf{x} = \mathbf{0}$ has only trivial solution, that is

   $$x_1\mathbf{c}_1 + \cdots + x_m\mathbf{c}_m = \mathbf{0}$$

   if and only if $x_1 = \ldots x_m - 0$ where $\mathbf{c}_1, \ldots, \mathbf{c}_m$ are the columns of A. This happens if and only if columns of A are independent

Problem

Show that the transformation defined by

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

is one-to-one.

Problem

Show that the transformation defined by

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

is one-to-one.

Solution

Since T is a matrix transformation induced by $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, and A has full column rank (=2), it follows that $A\mathbf{x} = 0$ has the unique solution $\mathbf{x} = 0$.

Problem

Show that the transformation defined by

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

is one-to-one.

Solution

Since T is a matrix transformation induced by $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, and A has full column rank (=2), it follows that $A\mathbf{x} = 0$ has the unique solution $\mathbf{x} = 0$. From the previous theorem we conclude that T is one-to-one.

Solution (cont)

Alternatively, observe that

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 1 & 0 \end{array}\right]$$

Solution (cont)

Alternatively, observe that

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 1 & 0 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right]$$

has unique solution $\boldsymbol{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, *so* T *is a one-to-one.*

Solution (cont)

Alternatively, observe that

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 1 & 0 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right]$$

has unique solution $\boldsymbol{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, *so* T *is a one-to-one. (Any other ways?)*

Example

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation defined by

$$T\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a+b \\ 0 \end{bmatrix} \text{ for all } \begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2.$$

Example

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation defined by

$$T\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a+b \\ 0 \end{bmatrix} \text{ for all } \begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2.$$

Then T is not onto.

Example

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation defined by

$$T\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a+b \\ 0 \end{bmatrix} \text{ for all } \begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2.$$

Then T is not onto. To see why, choose $\mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \in \mathbb{R}^2$.

Example

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation defined by

$$T\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a+b \\ 0 \end{bmatrix} \text{ for all } \begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2.$$

Then T is not onto. To see why, choose $\mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \in \mathbb{R}^2$. Then there is no vector $\mathbf{x} \in \mathbb{R}^2$ so that $T(\mathbf{x}) = \mathbf{b}$; applying T to any vector results in a vector whose second entry is 0, and the second entry of $\mathbf{b}$ is 1.

Example (continued)

Consider the system $A\mathbf{x} = \mathbf{b}$, where A is the matrix induced by T,

Example (continued)

Consider the system $A\mathbf{x} = \mathbf{b}$, where A is the matrix induced by T,

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Example (continued)

Consider the system $A\mathbf{x} = \mathbf{b}$, where A is the matrix induced by T,

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Then the augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right],$$

Example (continued)

Consider the system $A\mathbf{x} = \mathbf{b}$, where A is the matrix induced by T,

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Then the augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right],$$

which is already in reduced row-echelon form.

Example (continued)

Consider the system $A\mathbf{x} = \mathbf{b}$, where A is the matrix induced by T,

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Then the augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right],$$

which is already in reduced row-echelon form. The fact that this system is inconsistent implies that T is not onto.

Problem

Show that the transformation defined by

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

is onto.

Problem

Show that the transformation defined by

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

is onto.

Solution

T is a matrix transformation, $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$.

Problem

Show that the transformation defined by

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

is onto.

Solution

T is a matrix transformation, $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$.

Since $\det(A) = -1 \neq 0$, A is invertible.

Problem

Show that the transformation defined by

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

is onto.

Solution

T is a matrix transformation, $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$.

Since $\det(A) = -1 \neq 0$, A is invertible.
Therefore, for every choice of $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has an unique solution (namely $\mathbf{x} = A^{-1}\mathbf{b}$) . So T is onto.

Problem

Show that the transformation defined by

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

is onto.

Problem

Show that the transformation defined by

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

is onto.

Solution

T is a matrix transformation, $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$.

Problem

Show that the transformation defined by

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

is onto.

Solution

T is a matrix transformation, $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$.

Since $\det(A) = -1 \neq 0$, A is invertible.

Problem

Show that the transformation defined by

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

is onto.

Solution

T is a matrix transformation, $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$.

Since $\det(A) = -1 \neq 0$, A is invertible.
Therefore, for every choice of $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has an unique solution (namely $\mathbf{x} = A^{-1}\mathbf{b}$) . So T is onto.

One-to-one and onto

Problem

Let T be the linear transformation induced by $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$. Show that T_A is one-to-one and onto.

One-to-one and onto

Problem

Let T be the linear transformation induced by $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$. Show that T_A is one-to-one and onto.

Solution

Let R be a row-echelon form of A.

$$A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = R$$

In this case, A is invertible, so $A\mathbf{x} = \mathbf{b}$ has a unique solution $\mathbf{x}$ for every $\mathbf{b}$ in $\mathbb{R}^2$. Therefore T_A is both one-to-one and onto.

Neither one-to-one nor onto

Problem

Let T be the linear transformation induced by $A = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$.

Show that T_A is neither one-to-one nor onto.

Solution

Let R be a row-echelon form of A.

$$A = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} = R$$

Since A has rank two, the augmented matrix $[A|\mathbf{b}]$ will have rank three for some choice of $\mathbf{b} \in \mathbb{R}^3$, resulting in $A\mathbf{x} = \mathbf{b}$ being inconsistent. Therefore, T_A is not onto.

Solution

Let R be a row-echelon form of A.

$$A = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} = R$$

Since A has rank two, the augmented matrix $[A|\mathbf{b}]$ will have rank three for some choice of $\mathbf{b} \in \mathbb{R}^3$, resulting in $A\mathbf{x} = \mathbf{b}$ being inconsistent. Therefore, T_A is not onto.

The augmented matrix $[A|0]$ has rank two, so the system $A\mathbf{x} = 0$ has a non-leading variable, and hence does not have unique solution $\mathbf{x} = 0$. Therefore, T_A is not one-to-one.

Matrix of a One to One or Onto Transformation

Theorem

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation induced by the $m \times n$ matrix A. Then

1. *T is one to one if and only if the rank of A is n.*
2. *T is onto if and only if the rank of A is m.*

Exercise

Let

$$T : \mathbb{R}^2 \longrightarrow \mathbb{R}^3$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 3x_1 + x_2 \\ 5x_1 + 7x_2 \\ x_1 + 3x_2 \end{bmatrix}$$

Show that T is a one-to-one linear transformation. Is T onto $\mathbb{R}^3$?

Isomorphism

Definition

A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is said to be **isomorphism** if and only if it is both one-to-one and onto

Theorem

The linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ induced by the matrix A is isomorphism if and only if columns of A is are linear independent and span $\mathbb{R}^m$

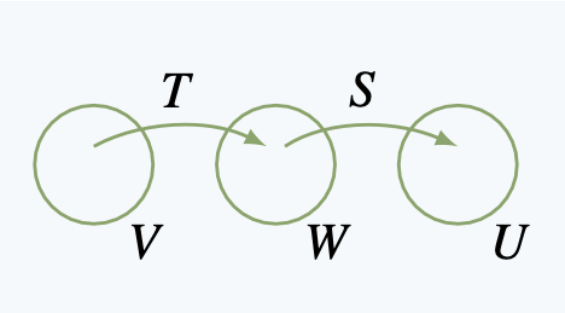
Composition of Linear Transformation

Given linear transformations $T : V \to W$ and $S : W \to U$, the composite

$$ST : V \to U$$

of T and S is defined by

$$(ST)(\mathbf{v}) = S(T(\mathbf{v}))$$



If T is induced by matrix A and S is induce by matrix B then ST is a linear transformation induced by matrix BA

Example

Determine composition of reflection about x - axis and contraction from $\mathbb{R}^2$ to $\mathbb{R}^2$ by a

Solution

Relfection about x - axis

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ -y \end{bmatrix}$$

Contraction transformation

$$S\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax \\ ay \end{bmatrix}$$

Composition of reflection about x-axis and contraction

$$(ST)\begin{bmatrix} x \\ y \end{bmatrix} = S\left(T\begin{bmatrix} x \\ y \end{bmatrix}\right) = S\begin{bmatrix} x \\ -y \end{bmatrix} = \begin{bmatrix} ax \\ -ay \end{bmatrix}$$

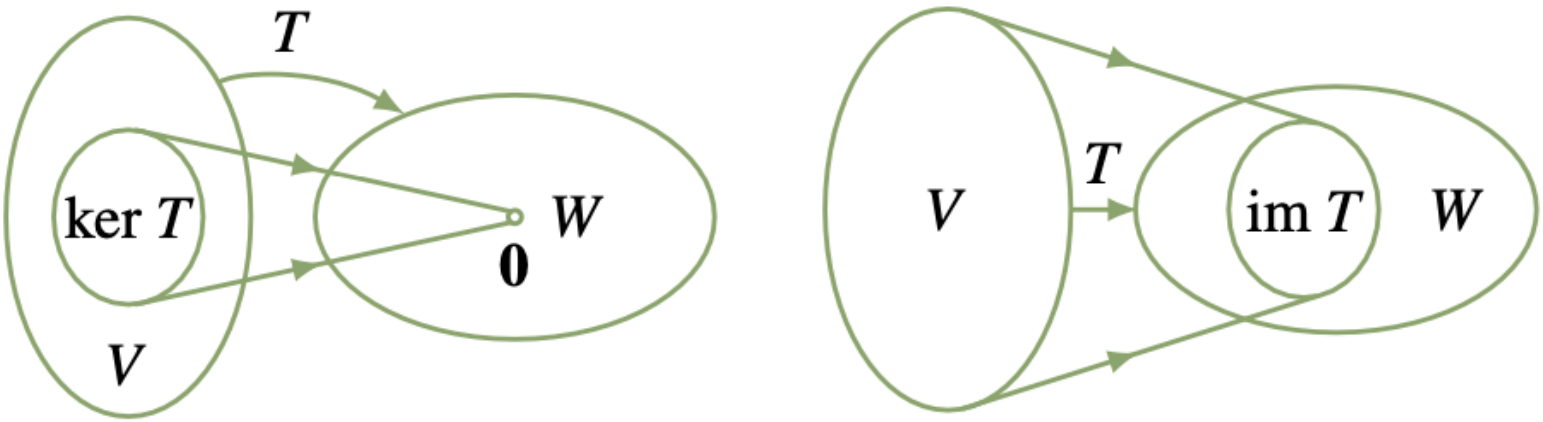
Exercise

Show that reflection in the x axis followed by rotation through $\frac{\pi}{2}$ is reflection in the line $y = x$.

Definition (Kernel and Image)

Let $T : V \to W$ be a linear transformation. The **kernel** of T (denote by $ker(T)$) and the **image** of T (denoted by $im(T)$ or $T(V)$) are defined by

$$ker(T) = \{\mathbf{v} \in V | T(\mathbf{v} = 0)\}$$
$$im(T) = \{T(\mathbf{v}|\mathbf{v} \in V) = T(V)\}$$



The kernel of T is often called the **nullspace** of T because it consists of all vectors $\mathbf{v}$ in V satisfying the condition that $T(\mathbf{v}) = 0$. The image of T is often called the **range** of T and consists of all vectors $\mathbf{w}$ in W of the form $\mathbf{w} = T(\mathbf{v})$ for some $\mathbf{v}$ in V

Example

Let $T : \mathbb{R}^3 \mapsto \mathbb{R}^2$ be defined by

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} a+b+c \\ c-a \end{bmatrix}$$

Then T is a linear transformation. Find a basis for $ker(T)$ and $im(T)$.

Example

Let $T : \mathbb{R}^3 \mapsto \mathbb{R}^2$ be defined by

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} a+b+c \\ c-a \end{bmatrix}$$

Then T is a linear transformation. Find a basis for $ker(T)$ and $im(T)$.

Solution

You can (and should!) verify that T is a linear transformation.

Solution (continued)

Kernel of T: We look for all vectors $\mathbf{x} \in \mathbb{R}^3$ such that $T(\mathbf{x}) = \mathbf{0}$.

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} a+b+c \\ c-a \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Solution (continued)

Kernel of T: We look for all vectors $\mathbf{x} \in \mathbb{R}^3$ such that $T(\mathbf{x}) = \mathbf{0}$.

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} a+b+c \\ c-a \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives a system of equations:

$$\begin{aligned} a+b+c &= 0 \\ c-a &= 0 \end{aligned}$$

Solution (continued)

Kernel of T: We look for all vectors $\mathbf{x} \in \mathbb{R}^3$ such that $T(\mathbf{x}) = \mathbf{0}$.

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} a+b+c \\ c-a \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives a system of equations:

$$\begin{aligned} a+b+c &= 0 \\ c-a &= 0 \end{aligned}$$

The general solution is

$$\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \left\{\begin{bmatrix} t \\ -2t \\ t \end{bmatrix} : t \in \mathbb{R}\right\} = \left\{t\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} : t \in \mathbb{R}\right\}$$

And therefore a basis for the kernel is $\left\{\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}\right\}$.

Solution (continued)

Image of T: We can write the image as

$$\begin{aligned} im(T) &= \left\{ \begin{bmatrix} a+b+c \\ c-a \end{bmatrix} : a,b,c \in \mathbb{R} \right\} \\ &= \left\{ \begin{bmatrix} a \\ -a \end{bmatrix} + \begin{bmatrix} b \\ 0 \end{bmatrix} + \begin{bmatrix} c \\ c \end{bmatrix} : a,b,c \in \mathbb{R} \right\} \\ &= \left\{ a \begin{bmatrix} 1 \\ -1 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \end{bmatrix} : a,b,c \in \mathbb{R} \right\} \end{aligned}$$

Solution (continued)

Image of T: We can write the image as

$$\begin{aligned} im(T) &= \left\{ \begin{bmatrix} a+b+c \\ c-a \end{bmatrix} : a,b,c \in \mathbb{R} \right\} \\ &= \left\{ \begin{bmatrix} a \\ -a \end{bmatrix} + \begin{bmatrix} b \\ 0 \end{bmatrix} + \begin{bmatrix} c \\ c \end{bmatrix} : a,b,c \in \mathbb{R} \right\} \\ &= \left\{ a \begin{bmatrix} 1 \\ -1 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \end{bmatrix} : a,b,c \in \mathbb{R} \right\} \end{aligned}$$

Thus $im(T) = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$.

Solution (continued)

Image of T: We can write the image as

$$\begin{aligned} im(T) &= \left\{ \begin{bmatrix} a+b+c \\ c-a \end{bmatrix} : a,b,c \in \mathbb{R} \right\} \\ &= \left\{ \begin{bmatrix} a \\ -a \end{bmatrix} + \begin{bmatrix} b \\ 0 \end{bmatrix} + \begin{bmatrix} c \\ c \end{bmatrix} : a,b,c \in \mathbb{R} \right\} \\ &= \left\{ a \begin{bmatrix} 1 \\ -1 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \end{bmatrix} : a,b,c \in \mathbb{R} \right\} \end{aligned}$$

Thus $im(T) = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$.

These vectors are not linearly independent, but the first two are so a basis for the image of T is

$$\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}.$$

Kernel and One to One

The kernel of a linear transformation gives important information about whether the transformation is one to one. Recall that a linear transformation T is one to one if and only if $T(\mathbf{x}) = \mathbf{0}$ implies $\mathbf{x} = \mathbf{0}$.

Kernel and One to One

The kernel of a linear transformation gives important information about whether the transformation is one to one. Recall that a linear transformation T is one to one if and only if $T(\mathbf{x}) = \mathbf{0}$ implies $\mathbf{x} = \mathbf{0}$.

Theorem

Let $T : V \mapsto W$ be a linear transformation where V is a subspace of $\mathbb{R}^n$ and W a subspace of $\mathbb{R}^m$.
Then T is one to one if and only if $\ker(T) = \{\boldsymbol{0}\}$.

Table of contents

1. Linear Transformation
   - Linear Transformation
   - Geometric Linear Transform of $\mathbb{R}^2$
2. Kernel and Image of Linear Transformation
   - One-to-one and Onto Transformation
   - Isomorphism and Composition
   - Kernel and Image

Dimension of the Kernel and Image

Theorem (Dimension Theorem)

Let $T : V \mapsto W$ be a linear transformation where V is a subspace of $\mathbb{R}^n$ and W is a subspace of $\mathbb{R}^m$. Suppose further that the dimension of V is k. Then

$$k = \dim(\ker(T)) + \dim(im(T))$$

Example (Revisited)

Let $T : \mathbb{R}^3 \mapsto \mathbb{R}^2$ be defined by

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} a+b+c \\ c-a \end{bmatrix}$$

Find the dimension of $\ker(T)$ and $im(T)$.

Example (Revisited)

Let $T : \mathbb{R}^3 \mapsto \mathbb{R}^2$ be defined by

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} a+b+c \\ c-a \end{bmatrix}$$

Find the dimension of $\ker(T)$ and $im(T)$.

Solution

We already know that a basis for the kernel of T is given by

$$\left\{\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}\right\}$$

Example (Revisited)

Let $T : \mathbb{R}^3 \mapsto \mathbb{R}^2$ be defined by

$$T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} a+b+c \\ c-a \end{bmatrix}$$

Find the dimension of $\ker(T)$ and $im(T)$.

Solution

We already know that a basis for the kernel of T is given by

$$\left\{\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}\right\}$$

Therefore $\dim(\ker(T)) = 1$.

Solution (continued)

We also found a basis for the image of T as

$$\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

and this of course shows $\dim(im(T)) = 2$.

Solution (continued)

We also found a basis for the image of T as

$$\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

and this of course shows $\dim(im(T)) = 2$.

But we could have found the dimension of $im(T)$ without finding a basis. That's because since the dimension of $\mathbb{R}^3$ is 3, and the dimension of $\ker(T)$ is 1, we get by the Dimension Theorem that:

$$\begin{aligned} \dim(im(T)) &= \dim(\mathbb{R}^3) - \dim(ker(T)) \\ &= 3 - 1 = 2 \end{aligned}$$

Nullity and Rank

Let

$$T_A : \mathbb{R}^n \ \rightarrow \mathbb{R}^m$$

be the linear transformation induced by the $m \times n$ matrix A, that is $T_A(\mathbf{x} = A\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$ then

$$ker(T_A) = \{\mathbf{x}|A\mathbf{x} \ = 0\} = null(A)$$
$$im(T_A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} = im(A)$$

which are subspace in $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively.

$dim(ker(T))$ is called the **nullity** of T and denoted as $nullity(T)$

$dim(im(T))$ is called the **rank** of T and denoted as $rank(T)$

Example

Given a $m \times n$ matrix A, show that $im(T_A) = col(A)$ and so $rank(T_A) = rank(A)$

Example

Given a $m \times n$ matrix A, show that $im(T_A) = col(A)$ and so $rank(T_A) = rank(A)$

Solution

Write $A = \begin{bmatrix} \boldsymbol{c}_1 & \cdots & \boldsymbol{c}_n \end{bmatrix}$ *in terms of its columns then*

$$im(T_A) = \{A\boldsymbol{x} | \boldsymbol{x} \in \mathbb{R}^n\} = \{x_1\boldsymbol{c}_1 + \cdots + x_n\boldsymbol{c}_n : x_i \in \mathbb{R}\} = col(A)$$